Guided Notes: Support Vector Machines, Margins, Kernels, and Regression — Complete

This version supplies worked responses to every student task.

These notes connect the geometry of a separating boundary to the optimization problem used by a support vector classifier. We then use kernels to make nonlinear boundaries and extend the same ideas to regression.

Learning goals

By the end of these notes, you should be able to:

1. interpret a linear decision score and calculate distance to its hyperplane;
2. explain why the hard-margin objective produces the widest separating margin;
3. calculate hinge loss and explain what the soft-margin parameter C does;
4. use a feature map, kernel, and support vectors to explain nonlinear prediction;
5. describe how C and the RBF parameter γ should be selected; and
6. calculate ε-insensitive loss for support vector regression (SVR).

1. A score, a boundary, and a distance

For binary classification, let $x_i \in \mathbb{R}^p$ be the features of training observation i and $y_i \in \{-1, +1\}$ its label. A linear classifier uses

$$f(x) = w^\top x + b, \qquad w \in \mathbb{R}^p, \quad b \in \mathbb{R}.$$

The decision boundary is the hyperplane $f(x) = 0$. For $p = 2$ it is a line. The nonzero vector w is perpendicular to the boundary. We predict $+1$ when $f(x) > 0$ and -1 when $f(x) < 0$; a score of zero needs a tie convention.

A. Read the score

Use $w = (3, 4)^\top$ and $b = -5$.

Point x	$w^\top x + b$	Predicted class
$(2, 1)^\top$	$3(2) + 4(1) - 5 = 5$	$+1$
$(1, 0)^\top$	$3(1) + 4(0) - 5 = -2$	-1

What does the sign tell us? Does a score of 5 mean a probability of 0.5?

The sign tells us which side of the boundary contains the point. A score of 5 is not a probability; it is an uncalibrated value of the fitted decision function. In particular, it does not mean 0.5 or 50%.

B. Turn a score into a perpendicular distance

The length of w is $\|w\|_2 = \sqrt{w^\top w}$. Its unit direction is $u = w/\|w\|_2$. Choose any point x_0 on the boundary, so $w^\top x_0 + b = 0$. Project $x - x_0$ onto u:

$$\begin{aligned}(x - x_0)^\top u &= \frac{w^\top x - w^\top x_0}{\|w\|_2} \\ &= \frac{w^\top x + b}{\|w\|_2},\end{aligned}$$

This is a **signed** distance. The ordinary perpendicular distance is

$$d(x, \{f = 0\}) = \frac{|w^\top x + b|}{\|w\|_2}.$$

For $w = (3, 4)^\top$, $b = -5$, and $x = (2, 1)^\top$, compute $\|w\|_2$, the score, and the distance.

$$\|w\|_2 = \sqrt{3^2 + 4^2} = 5, \quad f(x) = 3(2) + 4(1) - 5 = 5, \quad d = \frac{|5|}{5} = 1.$$

If we replace (w, b) by $(2w, 2b)$, what happens to the boundary, score, and distance?

The equation $2w^\top x + 2b = 0$ has the same solutions as $w^\top x + b = 0$, so the boundary and predictions are unchanged. Every score doubles, while the norm in the denominator also doubles: $|2f(x)|/\|2w\|_2 = |f(x)|/\|w\|_2$. Thus the geometric distance is unchanged. This is why raw score magnitudes cannot be compared across differently scaled parameterizations.

2. Choose the widest separating margin

The **margin** is the smallest perpendicular distance from a correctly classified training point to the boundary. A maximal-margin classifier chooses a separating boundary with the largest margin. The training points nearest the boundary are its **support vectors**.

For a small example, use $w = (1, 0)^\top$, $b = 0$, and the points below. All points have second coordinate zero.

ID	x_i	y_i	Score $f(x_i)$	Label-adjusted score $m_i = y_i f(x_i)$	Distance to boundary
A	$(-2, 0)^\top$	-1	-2	2	2
B	$(-1, 0)^\top$	-1	-1	1	1
C	$(1, 0)^\top$	$+1$	1	1	1
D	$(2, 0)^\top$	$+1$	2	2	2

1. Which points are support vectors for this boundary? What is the one-sided margin?
   B and C have the smallest distance, 1, so they are support vectors and the one-sided margin is 1.
2. What are the two margin-edge equations? What is the full distance between those edges?
   The edges are $f(x) = -1$ and $f(x) = +1$, or $x_1 = -1$ and $x_1 = +1$. They are 2 units apart. The decision boundary $x_1 = 0$ lies halfway between them.

A. Fix the arbitrary scale

Multiplying both w and b by any positive constant keeps the same boundary and predictions. For separable data, we choose the scale so the smallest label-adjusted score is 1:

$$\min_i y_i(w^\top x_i + b) = 1.$$

Suppose an initial separator $(\tilde{w}, \tilde{b})$ has smallest label-adjusted score 2. By what positive factor should we multiply both parameters to make that smallest score 1?

$$\text{Scale factor} = \frac{1}{2}.$$

More generally, if the smallest label-adjusted score is $a > 0$, set $(w, b) = (\tilde{w}/a, \tilde{b}/a)$. The boundary and predictions stay the same while the smallest new score becomes 1.

Then every training point satisfies $y_i(w^\top x_i + b) \geq 1$. The nearest points have distance $1/\|w\|_2$ and the distance between the two margin edges is $2/\|w\|_2$.

Complete the hard-margin optimization problem:

$$\boxed{\begin{aligned} &\underset{w,b}{\text{minimize}} && \frac{1}{2}\|w\|_2^2 \\ &\text{subject to} && y_i(w^\top x_i + b) \geq 1 \quad \text{for every } i. \end{aligned}}$$

Why does making $\|w\|_2$ smaller make the geometric margin wider **after** we have fixed the scale? Why would minimizing $\|w\|_2$ without the constraints fail?

The normalized nearest-point score is 1, so the one-sided geometric margin is $1/\|w\|_2$. Its reciprocal increases as $\|w\|_2$ decreases. The constraints ensure the boundary separates the classes and fix the otherwise arbitrary scale. Without them, $w = 0$ minimizes the norm but does not provide a meaningful separating direction.

3. When perfect separation is too strict

If the classes overlap, no hard-margin separator exists. Even when they are separable, demanding perfect training separation may give a boundary that responds too strongly to a few unusual observations.

For each point define the label-adjusted score $m_i = y_i f(x_i)$. The **hinge loss** is

$$L_{\text{hinge},i} = \max(0, 1 - m_i).$$

It is zero on or beyond the correct margin edge, positive inside the margin, and greater than 1 for a point on the wrong side of the decision boundary.

A. Locate a point from its margin score

Fill in the hinge loss and describe the point's location. At $m_i = 0$, the point lies exactly on the decision boundary.

m_i	Hinge loss	Location and classification
1.4	0	Correctly classified, beyond the margin edge

m_i	Hinge loss	Location and classification
1	0	Correctly classified, on the margin edge
0.4	0.6	Correctly classified, inside the margin
0	1	On the decision boundary; class needs a tie rule
-0.6	1.6	On the wrong side of the boundary; misclassified

Which row is correctly classified but still incurs a positive loss? Why?

The $m_i = 0.4$ row. Its score has the correct sign, but its label-adjusted score is below the desired margin value 1. Hinge loss penalizes margin violations, including correctly classified points inside the margin.

B. Trade margin width against violations

The soft-margin classifier introduces $\xi_i \geq 0$ and solves

$$\begin{array}{ll} \underset{w,b,\xi}{\text{minimize}} & \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\xi_i \\ \text{subject to} & y_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0. \end{array}$$

Here $C > 0$ is the **penalty per unit of slack**. For fixed w, b, the smallest feasible slack is

$$\xi_i = \max\{0, 1 - y_i(w^\top x_i + b)\},$$

so the objective can also be written as

$$\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\max\{0, 1 - y_i(w^\top x_i + b)\}.$$

Choice	How costly is a violation?	Typical pressure on the fitted boundary
Smaller C	Less costly	Tends to allow more margin violations and favor a wider, smoother margin
Larger C	More costly	Presses harder to avoid violations, often giving a narrower margin

Why is ξ_i measured in score units rather than Euclidean distance units? If $w \neq 0$, what is the geometric distance corresponding to a score shortfall ξ_i?

The constraint compares ξ_i directly with $1 - y_i f(x_i)$, which is a difference of label-adjusted **scores**. Division by $\|w\|_2$ converts that shortfall into the perpendicular distance from the point to its correct margin edge: $\xi_i / \|w\|_2$ when $\xi_i > 0$. A change in C changes a tradeoff; it does not prescribe a fixed margin width or a fixed number of support vectors.

A notation warning when reading ISLP: In its Section 9.2 theory, C names a *total slack budget*, so increasing that C allows more violations. These notes and scikit-learn use C as a *penalty coefficient*, so increasing it makes violations more costly. Check the objective before interpreting C.

4. Use the classification rule carefully

After fitting a linear classifier, predict from the sign of $f(x) = w^\top x + b$. For a fixed fitted model with $w \neq 0$, $|f(x)| / \|w\|_2$ measures distance to its boundary. The raw score is **not** a calibrated probability.

1. A new point has $f(x) = -2$. Which class does the sign rule predict?
   It predicts -1.
2. If all fitted coefficients and the intercept were multiplied by 3, what would happen to $f(x)$, the predicted class, and the distance to the boundary?
   The raw score would become -6. Its sign and predicted class would be unchanged. The perpendicular distance would also be unchanged because both $|f(x)|$ and $\|w\|_2$ triple.
3. Logistic regression and a linear SVM can both produce a linear boundary. What is one difference between their fitting losses or outputs?
   Logistic regression fits a Bernoulli probability model using log loss and outputs probabilities through the sigmoid. The soft-margin SVM uses hinge loss plus a norm penalty and directly outputs a decision score and class sign. An SVM score is not itself a probability.

5. From a linear boundary to a nonlinear one

A boundary can be linear in *transformed* features but nonlinear in the original features. Consider a one-feature problem with positive examples at $x = -2, 2$ and negative examples at $x = -0.5, 0.5$.

x	-2	-0.5	0.5	2
Label	$+1$	-1	-1	$+1$
$\phi(x) = x^2$	4	0.25	0.25	4

1. Why can one threshold on the original number line not separate these labels?
   The negative points are between the two positive points. A single threshold splits the number line into only a left and a right region, so it cannot put both outer points in one class and both inner points in the other.
2. In the transformed feature, use $f(x) = \phi(x) - 1 = x^2 - 1$. Fill in its four scores and state where its boundaries lie on the original number line.
   The scores, from left to right, are $3, -0.75, -0.75, 3$. Their signs classify all four examples correctly. Solving $x^2 - 1 = 0$ gives boundaries at $x = -1$ and $x = 1$ in the original feature space. One linear threshold in the transformed coordinate x^2 therefore becomes two boundary points on the original number line.

The corresponding kernel computes a transformed inner product without explicitly building the transformed features:

$$K(x, z) = \phi(x)\phi(z) = (xz)^2.$$

Calculate $K(2, -2)$ and $K(2, 0.5)$. Why can 2 and -2 be highly similar **under this kernel**?

$K(2, -2) = (2 \cdot -2)^2 = 16$, whereas $K(2, 0.5) = (2 \cdot 0.5)^2 = 1$. This kernel compares the squared feature: $\phi(2) = \phi(-2) = 4$. Opposite signs in the original coordinate disappear after squaring, so a large kernel value here reflects similarity in the transformed feature, not proximity on the original number line.

In this course, “SVM” can refer to the whole family. ISLP calls the linear soft-margin rule a **support vector classifier** and uses **support vector machine** for its kernel extension.

6. The kernel trick and the dual view

A valid kernel has the form $K(x, z) = \phi(x)^\top \phi(z)$ for some feature map ϕ, even when we never construct $\phi(x)$ explicitly. Its matrix of training-pair values $K(x_i, x_j)$ is the **Gram matrix** and must be positive semidefinite; an arbitrary similarity function may not qualify.

For the soft-margin classifier, a dual formulation chooses one weight α_i per training observation:

$$\begin{aligned}
\underset{\alpha}{\text{maximize}} \quad & \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \\
\text{subject to} \quad & 0 \leq \alpha_i \leq C, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0.
\end{aligned}$$

You do not need to solve this optimization by hand. Its useful consequence is the fitted decision function:

$$f(x) = \sum_{i:\alpha_i > 0} \alpha_i y_i K(x_i, x) + b.$$

Only training observations with $\alpha_i > 0$ appear in this sum; these are support vectors in the fitted dual solution.

A. Predict from support vectors

Suppose there are two support vectors, $x_1 = -1$, $y_1 = -1$, $x_2 = 1$, $y_2 = +1$, with $\alpha_1 = \alpha_2 = 1/2$, $b = 0$, and linear kernel $K(u, v) = uv$. Complete

$$\begin{aligned}
f(x) &= \tfrac{1}{2}(-1)K(-1, x) + \tfrac{1}{2}(+1)K(1, x) \\
&= \tfrac{1}{2}(-1)(-x) + \tfrac{1}{2}(1)(x) = x.
\end{aligned}$$

What classes are predicted at $x = -2$ and $x = 3$?

$f(-2) = -2$, so the prediction is -1. $f(3) = 3$, so the prediction is $+1$.

B. Read a dual weight

At an exact soft-margin optimum, match each case to its typical geometric location. A point exactly on a margin edge can be degenerate, so a zero weight does not always prove strict distance beyond the margin.

Dual weight	What can we conclude about $m_i = y_i f(x_i)$?
$\alpha_i = 0$	$m_i \geq 1$; it can equal 1 in a degenerate solution
$0 < \alpha_i < C$	$m_i = 1$; the point is on its margin edge
$\alpha_i = C$	$m_i \leq 1$; the point is on or inside its margin edge

Does $\alpha_i = C$ necessarily mean the point is misclassified? Explain.

No. For example, $0 < m_i < 1$ is correctly classified but inside the margin, and such a violation has $\alpha_i = C$ at an exact optimum. Even $m_i = 1$ can occur with $\alpha_i = C$. Misclassification requires $m_i < 0$.

7. Common kernels and choosing their parameters

Kernel	Formula	Effect
Linear	$K(x, z) = x^\top z$	Linear boundary in original features
Polynomial	$K(x, z) = (\gamma x^\top z + r)^d$	Polynomial interactions, degree d
Gaussian RBF	$K(x, z) = \exp(-\gamma \lVert x - z \rVert_2^2)$	Local similarity, $\gamma > 0$

For a fixed nonzero distance $\lVert x - z \rVert_2^2 = 1$, compare $K(x, z)$ when $\gamma = 0.25$ and $\gamma = 2$. Which value of γ makes similarity fall off more rapidly with distance?

$$K_{\gamma=0.25} = e^{-0.25} \approx 0.779, \qquad K_{\gamma=2} = e^{-2} \approx 0.135.$$

The larger γ makes similarity fall off more rapidly for this fixed distance.

Explain why a larger γ often permits a more locally varying RBF boundary. Is a more flexible boundary guaranteed to perform better on new data?

With larger γ, a training point has substantial RBF similarity only to points quite near it. Individual support vectors can therefore affect smaller neighborhoods, often allowing the decision surface to vary more locally. This can fit training detail or noise, so performance on new data must be checked with validation rather than inferred from flexibility alone.

Model-selection check

A classmate standardizes all features using the full dataset, splits into training and test sets, tunes C and γ on the test set, and reports that test score. Identify two problems and describe a sound sequence instead.

First, full-dataset standardization leaks information from future test observations into training. Second, tuning on the test set makes the reported test score optimistic because the test set helped choose the model. Split off a test set first; within the training set, fit scaling separately inside each cross-validation training fold and tune C and γ there. Refit the selected pipeline on all training data, then evaluate once on the untouched test set. Choose a metric suited to the classification goal and class balance.

Remember that kernel training uses an $n \times n$ Gram matrix, so a nonlinear kernel can become expensive when n is large. A linear method may be more practical in that setting.

8. Support vector regression: an ε-wide tube

For regression, y_i is numeric and the prediction is a number $f(x)$, not a class sign. A linear SVR uses $f(x) = w^\top x + b$. The ε-tube contains observations with $|y_i - f(x_i)| \le \varepsilon$; its full **vertical** width is 2ε in response units.

The ε-insensitive loss is

$$L_\varepsilon(y_i, f(x_i)) = \max\{0,\ |y_i - f(x_i)| - \varepsilon\}.$$

With $\varepsilon = 0.5$, complete the table.

Residual $r_i = y_i - f(x_i)$	$\lvert r_i \rvert$	L_ε	**Inside, on, or outside the tube?**
-0.2	0.2	0	Inside
0.5	0.5	0	On the edge
0.8	0.8	0.3	Outside, above the fitted value
-1.1	1.1	0.6	Outside, below the fitted value

Why is the loss zero for the first row even though the prediction is not exact?

SVR deliberately ignores residual magnitudes up to $\varepsilon = 0.5$. The first residual has magnitude $0.2 < 0.5$, so $\max(0, 0.2 - 0.5) = 0$.

An equivalent SVR objective is

$$\underset{w,b}{\text{minimize}} \quad \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \max\{0,\ |y_i - f(x_i)| - \varepsilon\}.$$

The slack form uses one variable for each side of the tube:

$$
\begin{aligned}
\underset{w,b,\xi,\xi^*}{\text{minimize}} \quad & \frac{1}{2}\|w\|_2^2 + C\sum_i(\xi_i + \xi_i^*) \\
\text{subject to} \quad & y_i - f(x_i) \le \varepsilon + \xi_i, \\
& f(x_i) - y_i \le \varepsilon + \xi_i^*, \qquad \xi_i, \xi_i^* \ge 0.
\end{aligned}
$$

Which slack applies when the observed response is **above** the tube? What does increasing ε change, and what does increasing C change?

If $y_i > f(x_i) + \varepsilon$, then $y_i - f(x_i) > \varepsilon$ and ξ_i measures the excess above the tube. Increasing ε widens the zero-loss tube in response units. Increasing C makes each unit of excess residual more costly relative to the norm penalty. As in classification, the fitted response can change in ways that depend on the data; these parameters do not set a fixed number of support vectors.

With a kernel, the SVR prediction has the form

$$
f(x) = \sum_i(\alpha_i - \alpha_i^*)K(x_i, x) + b.
$$

Points strictly inside the tube have zero dual weights. Points on or outside the tube can be support vectors. Why do we use the numeric value of $f(x)$ rather than its sign?

SVR predicts a quantitative outcome in the units of y. The sign would discard the magnitude and turn the regression output into a binary label.

9. Put the methods together

Question	Support vector classification	Support vector regression
Type of outcome	Binary label -1 or $+1$	Numeric response
Prediction	$\text{sign}(f(x))$	Numeric $f(x)$
Loss from a point	$\max(0, 1 - y_i f(x_i))$	$\max(0, \lvert y_i - f(x_i)\rvert - \varepsilon)$
Main tuning choices	C, kernel, and kernel parameters such as γ	C, ε, kernel, and kernel parameters

The formulations here use two classes. For more classes, software can combine binary SVMs, for example through one-versus-one or one-versus-rest classification.

Exit questions

1. A correctly classified point has $m_i = 0.3$. Is it beyond the margin, inside it, or misclassified? What is its hinge loss?
   It is correctly classified because $m_i > 0$, but inside its margin because $m_i < 1$. Its hinge loss is $1 - 0.3 = 0.7$.
2. Why does multiplying (w, b) by a positive constant preserve the geometric boundary and distance but change the raw score?
   Both $f(x) = 0$ and $cf(x) = 0$ describe the same boundary for $c > 0$, while the sign stays the same. The score becomes $cf(x)$, but the distance is $|cf(x)|/\|cw\|_2 = |f(x)|/\|w\|_2$.
3. What does a kernel replace in the dual optimization and prediction rule?
   It replaces the inner product $\phi(x_i)^\top \phi(x_j)$ between transformed training points in the dual objective, and $\phi(x_i)^\top \phi(x)$ in prediction. We evaluate K directly without constructing the transformed vectors.
4. In SVR with $\varepsilon = 0.4$, what loss does an absolute residual of 0.9 incur?
   The ε-insensitive loss is $\max(0, 0.9 - 0.4) = 0.5$.

Sources and further reading

- STAT 486, *Support Vector Machines*, lecture deck 12 (especially slides 3–24, 29–48, and 49–54).
- James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning with Applications in Python* (ISLP), Chapter 9: §9.1 on separating hyperplanes and margins, §9.2 on soft margins, §9.3 on kernels, and §9.5 on hinge loss and the regression extension. ISLP's theoretical C in §9.2 uses a different convention from the penalty C here.
- scikit-learn User Guide: Support Vector Machines, especially “Kernel functions” and “Mathematical formulation” for classification and SVR.